

EDUCATION

University of Washington (UW)

2022 - Present

Ph.D. Candidate in Computer Science and Engineering

Expected Grad. Date: Mar. 2027

Advisor: Prof. Byron Boots

Korea Advanced Institute of Science and Technology (KAIST)

2020 - 2022

M.S. in Artificial Intelligence

Advisor: Prof. Jaegul Choo

GPA: 4.06 / 4.30

Korea University

2013 - 2019

B.S. in Computer Science and Engineering

GPA: 3.70 / 4.50; Major GPA: 4.11 / 4.50

Military service during 2015 - 2016

RESEARCH INTEREST

Vision-Language to Action (VLA), Robot Perception, Perception for Model Predictive Control (MPC)

SELECTED PROJECTS

Agentic Visual Understanding for 3D Open-vocabulary Perception

Project lead

Jun. 2026 - Present

Agentic system for open-vocabulary object discovery and localization in 3D LiDAR + RGB

Keywords: Gemini, Gemma, Object Detection

Vision-Language to Navigation for Outdoor Autonomy: Complex/Unstructured Environments

Project lead

Oct. 2025 - Present

Language-guided planning and control with complex dynamics in unstructured environments

Keywords: Vision-language to Action (VLA), Model Predictive Control, Off-road, Husky

DARPA Robotic Autonomy in Complex Environments with Resiliency (RACER)

UW Perception Team

Sep. 2022 - Dec. 2025

High-speed ground vehicle autonomy in complex off-road terrain.

Keywords: Geometry Estimation, Uncertainty Estimation, BEV Segmentation

Uncertainty-aware Accurate Terrain Elevation Modeling

Project lead

Mar. 2025 - Jun. 2025

Predictive distribution modeling of terrain ground elevation with Neural Processes

Keywords: Ground Geometry, Uncertainty, Neural Processes, Bayesian Updates

Image-based Traversability Prediction using Self-supervision

Project Lead

Mar. 2023 - Jan. 2024

Visual traversability learning from self-supervision signals.

Keywords: Contrastive Learning, Learning from Demonstration, Visual Traversability

PEER-REVIEWED PUBLICATIONS

* denotes equal contributions

- [13] Tyler Han, Yanda Bao, Bhaumik Mehta, Gabriel Guo, **Sanghun Jung**, Anubhav Vishwakarma, Emily Kang, Rosario Scalise, Jason Liren Zhou, Bryan Xu, and Byron Boots. Model Predictive Adversarial Imitation Learning for Planning from Observation. *International Conference on Learning Representations (ICLR)*, 2026. [paper]
- [12] **Sanghun Jung**, Daehoon Gwak, Byron Boots, and James Hays. Uncertainty-aware Accurate Elevation Modeling for Off-road Navigation via Neural Processes. *Conference on Robot Learning (CoRL)*, 2025. [paper]

- [11] Tyler Han, Preet Shah, Sidharth Rajagopal, Yanda Bao, **Sanghun Jung**, Sidharth Talia, Gabriel Guo, Bryan Xu, Bhaumik Mehta, Emma Romig, Rosario Scalise, and Byron Boots. Wheeled Lab: Modern Sim2Real for Low-cost, Open-source Wheeled Robotics *Conference on Robot Learning (CoRL)*, 2025. [paper] [code]
- [10] **Sanghun Jung**, Jingjing Zheng, Ke Zhang, Nan Qiao, Albert Y. C. Chen, Lu Xia, Chi Liu, Yuyin Sun, Xiao Zeng, Hsiang-Wei Huang, Byron Boots, Min Sun, and Cheng-Hao Kuo. Detail Matters for Indoor Open-vocabulary 3D Instance Segmentation. *International Conference on Computer Vision (ICCV)*, 2025. [paper] [code]
- [9] Hsiang-Wei Huang, Fu-Chen Chen, Wenhao Chai, Che-Chun Su, Lu Xia, **Sanghun Jung**, Cheng-Yen Yang, Jenq-Neng Hwang, Min Sun, and Cheng-Hao Kuo. Zero-shot 3D Question Answering via Voxel-based Dynamic Token Compression. *Computer Vision and Pattern Recognition (CVPR)*, 2025. [paper]
- [8] Xiangyun Meng, Xuning Yang, **Sanghun Jung**, Fabio Ramos, Srid Sadhan Jujjavarapu, Sanjoy Paul, and Dieter Fox. Aim My Robot: Precision Local Navigation to Any Object. *Robotics and Automation Letters (RA-L)*, 2025. [paper]
- [7] **Sanghun Jung**, JoonHo Lee, Xiangyun Meng, Byron Boots, and Alexander Lambert. V-STRONG: Visual Self-Supervised Traversability Learning for Off-road Navigation. *International Conference on Robotics and Automation (ICRA)*, 2024. [paper] [code]
- [6] Amirreza Shaban*, Brian JoonHo Lee*, **Sanghun Jung***, Xiangyun Meng, and Byron Boots. LiDAR-UDA: Self-ensembling Through Time for Unsupervised LiDAR Domain Adaptation. *International Conference on Computer Vision (ICCV)*, 2023. **Oral Presentation** (1.8% acceptance rate) [paper] [code]
- [5] **Sanghun Jung**, Jungsoo Lee, Nanhee Kim, Amirreza Shaban, Byron Boots, and Jaegul Choo. CAFA: Class-Aware Feature Alignment for Test-Time Adaptation. *International Conference on Computer Vision (ICCV)*, 2023. [paper]
- [4] Kyungmin Jo*, Gyumin Shim*, **Sanghun Jung**, Soyoung Yang, and Jaegul Choo. CG-NeRF: Conditional Generative Neural Radiance Fields. *Winter Conference on Applications of Computer Vision (WACV)*, 2023. [paper]
- [3] **Sanghun Jung***, Jungsoo Lee*, Daehoon Gwak, Sungha Choi, and Jaegul Choo. Standardized Max Logits: A Simple yet Effective Approach for Identifying Unexpected Road Obstacles in Urban-Scene Segmentation. *International Conference on Computer Vision (ICCV)*, 2021. **Oral Presentation** (3.0% acceptance rate) [paper] [code]
- [2] Sungha Choi*, **Sanghun Jung***, Huiwon Yun, Joanne T. Kim, Seungryong Kim, and Jaegul Choo. RobustNet: Improving Domain Generalization in Urban-Scene Segmentation via Instance Selective Whitening. *Computer Vision and Pattern Recognition (CVPR)*, 2021. **Oral Presentation** (4.1% acceptance rate) [paper] [code]
- [1] Jinho Choi, **Sanghun Jung**, Deokgun Park, Jaegul Choo, and Niklas Elmqvist. Visualizing for the Non-Visual: Enabling the Visually Impaired to Use Visualization. *Computer Graphics Forum (EuroVIS)*, 2019. [paper]

PREPRINTS

- [3] Tyler Han, Siyang Shen, Rohan Baijal, Harine Ravichandiran, Bat Nemekbold, Kevin Huang, **Sanghun Jung**, Byron Boots. Planning from Observation and Interaction. *Under Review*, 2026.
- [2] Jungsoo Lee, Juyoung Lee, **Sanghun Jung**, and Jaegul Choo. Improving Evaluation of Debiasing in Image Classification. *arXiv preprint: 2206.03680*, 2023. [paper]
- [1] Minsoo Lee, Chaeyeon Chung, Hojun Cho, Minjung Kim, **Sanghun Jung**, Minhyuk Sung, and Jaegul Choo. 3D-GIF: 3D-Controllable Object Generation via Implicit Factorized Representations with Unposed 2D Images. *arXiv preprint: 2203.06457*, 2022. [paper]

WORK EXPERIENCE

Waymo

Research Scientist Intern

Advancing perception foundation models for autonomous driving

Amazon Lab126

Applied Scientist Intern

Conducted research on VLA for improving robotic manipulation tasks

Mountain View, CA

Jun. 2026 - Present

Sunnyvale, CA

Jun. 2025 - Sep. 2025

Amazon Lab126

Bellevue, WA

Applied Scientist Intern

Jun. 2024 - Sep. 2024

Conducted research on open-vocabulary indoor 3D instance segmentation, published at ICCV'25

Bear Robotics

Redwood City, CA / Seoul, South Korea

Robotics Engineer Intern / Robotics Engineer

2018 - 2020

Conducted projects on velocity control, sensor calibration, and localization

SCHOLARSHIP**KAIST Support Scholarship**, KAIST

2020, 2021

Veritas Program Scholarship, Korea University

2018

Academic Excellence Scholarship for Freshmen, Korea University

2013

AWARDS**Best Poster Award - Standardized Max Logits**, KAIST AI Workshop

2022

PEER-REVIEW SERVICES**CVPR, ICRA ICCV**

2023

CVPR, IJCV, WACV

2024

CVPR, IROS, ICCV, CoRL, IJCV

2025

CVPR, AAAI, WACV, IJCV, RA-L, ECCV, BMVC, IJRR, NeurIPS, CoRL, TVCG,**T-FR, TPAMI**

2026

WACV, AAAI

2027

TEACHING**CSE 415 - Introduction to AI, Teaching Assistant**

Spring 2026

CSE 415 - Introduction to AI, Teaching Assistant

Fall 2026

INVITED TALKS**Pre-Training for Robot Learning Workshop @ CoRL 2023 (Spotlight Talk)**

Nov., 2023

Visual Self-Supervised Traversability Learning for Off-road Navigation

Hyundai Motor Group AI Research Seminar

Jul., 2021

Domain Generalization in Urban-Scene Segmentation

Naver AI LAB

Jul., 2021

RobustNet: Improving Domain Generalization in Segmentation